# HiVo

## Recording past and future audio on your phone

## 1. Problem

When you hear something that you want to save, for example, an interesting segment on the radio, a baby's first word, or a rare bird call, it is often too late to record it. Unless someone was recording, that audio is lost forever.

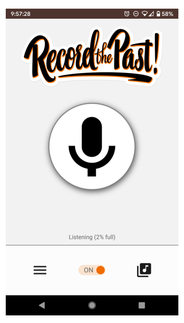
The aim of this project is to create a mobile solution that provides DVR-like functionality for audio on a mobile device.

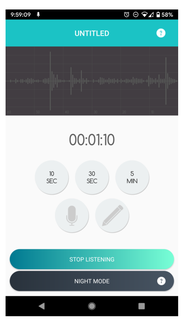
## 2. Existing solutions

There are several varied existing solutions to different use cases of this app, but none provide all the functionality of this project: keeping the last few hours recorded, and allowing the user to save a segment of it, along with scheduling recordings for the future.

There are several apps that constantly record audio, then allow the user to save some part of it. I analysed three.

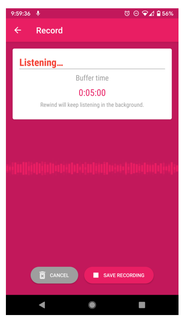
*Past Recorder* allows the user to save one of six set amounts (such as the last 30 seconds, or the last 15 minutes), up to 30 minutes, then crop and play back the result.

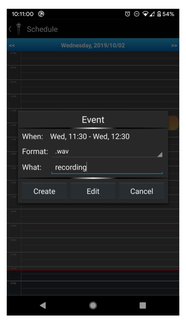



*Snipback* allows the user to save either one of three predefined amounts (which they can choose from eleven set amounts, up to 30 minutes, before they start recording), or the whole recording.

*Rewind* allows the user to save one of six set amounts, up to three hours, which they choose before they start recording.




CinixSoft's *Schedule Voice Recorder* app allows the user to schedule audio recordings.

## 3. Solution

This project addresses the project's aim using an Android phone's ability to record audio. After installing the HiVo app on their phone and turning it on (to "listen"), the app will store the last user-specified duration of audio. The user can then select segments of audio the device has listened to, saving them for future use. The user can also schedule recordings to start and end at a set time.

The HiVo app is compatible with the vast majority of Android devices, so it can be installed on an old phone rather than consuming the processing power and battery of the user's main device.

By default, it uses the phone's internal microphone. However, if there is an audio input device plugged into the phone, such as an FM radio or an external microphone, it uses that instead.

Download it now!
olliechick.co.nz/hivo

## 4. Design

The app has four screens. The main screen shows the audio-stream visualisation, and from here users can save chunks of audio. **Past Recordings** has a list of past recordings. **Scheduled Recordings** has a list of scheduled recordings, and gives users the ability to add more. **Settings** allows the user to set how far into the past the app keeps audio that was "listened to" before discarding it, and the filename format.

## 5. Implementation

When recording is started, an empty WAV file is created in internal storage with the WAV headers, then the audio data is added onto it. Amplitude data is sent to the main activity using intents, which renders it on the canvas by moving all existing lines left and adding the new amplitude to the very right. When recording is stopped, the two size headers are updated. To save all the audio, this file is copied into external storage. To save a section of audio, a new WAV file is created with the headers, and the relevant samples are copied over from the original file, then the size headers updated. This file is then moved into external storage.

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Chunk Descriptor ("RIFF") | | | | Chunk Size (Subchunk 2 Size + 36 = 64) | | | | Format ("WAVE") | | | |
| 52 | 49 | 46 | 46 | 40 | 00 | 00 | 00 | 57 | 41 | 56 | 45 |
| Subchunk 1 ID ("fmt ") | | | | Subchunk 1 size (16) | | | | Audio Format (1 for PCM) | | Number of Channels (2) | |
| 66 | 6d | 74 | 20 | 10 | 00 | 00 | 00 | 01 | 00 | 02 | 00 |
| Sample Rate (44100) | | | | Byte Rate (Sample Rate × Number of Channels × Bits per Sample / 8 = 176400) | | | | Block Align (Number of Channels × Bits per Sample / 8 = 4) | | Bits per Sample (16) | |
| 44 | ac | 00 | 00 | 10 | b1 | 02 | 00 | 04 | 00 | 10 | 00 |
| Subchunk 2 ID ("data") | | | | Subchunk 2 size (size of data = 28) | | | | Sample 0 (left) | | Sample 0 (right) | |
| 64 | 61 | 74 | 61 | 1c | 00 | 00 | 00 | 00 | 00 | 00 | 00 |
| Sample 1 (left) | | Sample 1 (right) | | Sample 2 (left) | | Sample 2 (right) | | Sample 3 (left) | | Sample 3 (right) | |
| 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 |
| Sample 4 (left) | | Sample 4 (right) | | Sample 5 (left) | | Sample 5 (right) | | Sample 6 (left) | | Sample 6 (right) | |
| 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 |

## 6. Future work

There are several more features that could be implemented. The biggest is the ability for users to play back the audio as the app is listening to it, scrubbing through it to find the segment they want to save. This would greatly improve the app's usability.

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